

Johns Hopkins University
Masters in Artificial Intelligence
Module 5 – Memory Assignment

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Program Description

The program implements CUDA kernels to convert a BMP image to grayscale. A testing image was randomly generated to avoid importing a file. The difference between the kernels is simply which memory type it is used to read the grayscale coefficients used in the calculation. The memory types are global memory, shared memory, constant memory, registers, and host memory. The host-memory kernel was implemented using mapped memory.

This is the formula used to convert from RGB to grayscale:

$$\text{Grayscale} = R * 0.299 + G * 0.587 + B * 0.114$$

R, G, and B are the red, green, and blue components of each BMP pixel.

Using the same grayscale coefficients across kernels was my attempt to fairly compare the performance of the different memory types. Constant memory is not writable, and registers are too small to hold an image larger than 18 x 18 pixels.

The program may take two optional and mutually exclusive parameters:

- threads: total number of threads per block (maximum is 1,024).
- num_blocks: total number of blocks desired.

If the “-threads” argument is provided, the number of blocks is calculated to approximate the image size, and vice versa. If the user doesn’t provide arguments, 256 threads per block, which translates to a tile size of 16 x 16. The implementation used a 2D grid.

The image size is predetermined as 7680 x 4320 (width x height).

Final Project

I intend to develop a Sobel filter for my final project, and converting a color image to grayscale is the first step.

The Sobel filter is one of the algorithms used to detect edges in images using the first derivative (image gradient). It applies two convolutions: one in the horizontal direction and one in the vertical direction.

Program Usage Examples:

#1 No Arguments Used

```
C:\Users\Herbert\workspace\EN605.617\module5>.\assignment.exe

Values Used:
Width  : 7680
Height : 4320
Image pixels: 33177600
Number of blocks: 129600
Threads per block: 256
Tile (Block) dimensions: (16 x 16)

Image: 7680 x 4320
Grid: (480, 270) Block: (16, 16)

Test Image Sample RGB values (RGB):
Test Image Sample RED pixels:  31 80 108 220 16
Test Image Sample GREEN pixels: 18 127 215 49 204
Test Image Sample BLUE pixels: 136 223 54 92 160

Average kernel time (3 warmup, 20 kernel runs, 500 loops inside kernel):
Global Mem      : 7.735 ms
Constant Mem    : 6.499 ms
Shared Mem      : 6.500 ms
Registers       : 6.504 ms
Host Mem        : 6.487 ms

Test Image Sample Grayscale values:
Test Image Sample GRAY pixels: 35 124 165 105 143
```

#2 Specify the Number of Threads per Block

```
C:\Users\Herbert\workspace\EN605.617\module5>.\assignment.exe -threads 1024

Values Used:
Width : 7680
Height : 4320
Image pixels: 33177600
Number of blocks: 32400
Threads per block: 1024
Tile (Block) dimensions: (32 x 32)

Image: 7680 x 4320
Grid: (240, 135) Block: (32, 32)

Test Image Sample RGB values (RGB):
Test Image Sample RED pixels: 87 79 189 249 42
Test Image Sample GREEN pixels: 60 201 126 90 12
Test Image Sample BLUE pixels: 133 170 5 60 222

Average kernel time (3 warmup, 20 kernel runs, 500 loops inside kernel):
Global Mem : 8.848 ms
Constant Mem : 6.869 ms
Shared Mem : 6.698 ms
Registers : 6.688 ms
Host Mem : 6.756 ms

Test Image Sample Grayscale values:
Test Image Sample GRAY pixels: 76 161 131 134 45
```

Specify the Number of Blocks

```
C:\Users\Herbert\workspace\EN605.617\module5>.\assignment.exe -num_blocks 120000

Values Used:
Width : 7680
Height : 4320
Image pixels: 33177600
Number of blocks: 111222
Threads per block: 256
Tile (Block) dimensions: (23 x 13)

Image: 7680 x 4320
Grid: (334, 333) Block: (23, 13)

Test Image Sample RGB values (RGB):
Test Image Sample RED pixels: 68 145 171 178 65
Test Image Sample GREEN pixels: 55 126 202 141 156
Test Image Sample BLUE pixels: 47 58 15 101 0

Average kernel time (3 warmup, 20 kernel runs, 500 loops inside kernel):
Global Mem : 8.780 ms
Constant Mem : 7.439 ms
Shared Mem : 7.029 ms
Registers : 7.021 ms
Host Mem : 7.052 ms

Test Image Sample Grayscale values:
Test Image Sample GRAY pixels: 58 124 171 148 111
```